

Essential Criteria

Qualifications, experience and knowledge

Possession of a PhD or equivalent qualification in a relevant subject area I hold a PhD in Electrical Engineering from the [University of Southern California](#), completed within its [Signal and Image Processing Institute](#).

An innovative personal research agenda in NeuroAI, computational neuroscience, or a A/I closely related field. Please refer to my Research Vision described in the attached rapelaVisionStmt.pdf.

Evidence of the ability to deliver and publish internationally excellent research. Publications should demonstrate both methodological rigour (e.g. development or application of quantitative/mathematical frameworks) and engagement with genuine biological or neuroscientific questions, not purely technical contributions to AI/ML alone. I have followed a non-traditional yet highly rigorous career path uniquely tailored to the modern demands of computational neuroscience. Following a PhD and two postdoctoral fellowships at the interface of statistical machine learning and neuroscience—resulting in key publications in leading journals such as *Neural Computation*, *Network: Computation in Neural Systems*, and the *Journal of Vision*—I chose to embed myself deeply within world-class experimental environments. I joined the Gatsby Computational Neuroscience Unit and the Sainsbury Wellcome Centre (SWC) as a Research Engineer Fellow to directly bridge the gap between advanced statistical methodology and world-class neuroscience experimentation.

At the Gatsby Unit and SWC I have grown enormously. I have learned world class machine learning at the Unit, and meeting one-on-one weekly with the Unit’s director, Maneesh Sahani, for more than seven years. I have acquired professional software development skills. I have been part of a unique computational/experimental neuroscience collaborative environment, where I actively participated in the institute-wide team science Aeon project. I was awarded £800.000 BBSRC award, and wrote a joint application between the Gatsby Unit, the SWC and the Allen Institute for Neural Dynamics for a BBSRC/NSF-Bio award. I lead a small team of research software engineers for more than three years. I designed and taught the Statistical Neuroscience course for SWC PhD students for two years. And I mentored two undergraduate students in one-year-long fellowships sponsored by the Simmons Foundation.

However, during this time my focus was on software development and ML training and not on research publications. Now I want to center my academic activities around publishing research articles, teaching and mentoring students, and collaborating with experimental groups at NPP and CDB. I am excellently prepared to do this. In the last seven years I have experimented with several paper ideas that will allow me to publish several high-quality papers in the next few years. Below I summarise a few of them:

- **Traveling waves during rhythmic speech production:** Expanding on single-subject findings reported in my earlier work (Rapela 2016, 2017, 2018), I am leveraging a newly acquired dataset across six subjects to publish a robust, high-impact characterization of speech-related cortical traveling waves.
- **svGPFA distribution and scientific papers:** I am preparing a pair of manuscripts documenting the unique numerical and software engineering architecture of our Python distribution of svGPFA, alongside a scientific paper applying the method to detect complex latent structures, such as replay in the striatum.
- **svGPFA extensions:** I am extending the core svGPFA framework to incorporate external latent drivers and real-time, online operation.
- **Bonsai.ML and Aeon data processing papers:** I am leading the writing of journal articles that introduce the *Bonsai.ML* package for intelligent experimental control and, in collaboration with an exceptional undergraduate mentee, reporting novel statistical insights from the *Aeon* project.
- **State-space models for behavioral inference:** To address the failure of deep pose-estimation networks to account for temporal continuity, we have developed [state-space models to infer kinematics](#). I will describe these methods and demonstrate their advantages in capturing multi-animal behavioral continuity.

Evidence of potential to secure competitive research funding. During my postdoctoral position at UC San Diego I was awarded a seed grant from *Center for Brain Activity Mapping* and at the Gatsby Unit an £800.000 grant from BBSRC.

Evidence of effective supervision and mentoring of undergraduate, postgraduate and/or doctoral research students including teaching in a computational, data science, or quantitative context (e.g. programming, machine learning, statistics, mathematical modelling) — ideally in neuroscience or bioscience settings; Python experience strongly preferred. As part of the [Simons Foundation’s SURFiN program](#) I mentored an undergraduate student from QMUL and another from UCL’s NPP.

I have been teaching assistant, discussion leader, course coordinator and instructor in courses in Python programming artificial intelligence, numerical linear algebra, probability, statistical neuroscience, delivered to undergraduate and graduate students. Please refer to my CV for details.

Evidence of productive cross-disciplinary collaborations, specifically spanning computational or AI methods and neuroscience or biological systems, for example joint A/I publications, grants, or collaborative projects with researchers from complementary disciplines. I owe much of my research trajectory in computational neuroscience to close partnerships with experimental collaborators:

- During my PhD, I collaborated with Dr. John Touryan and Dr. Gidon Felsen (UC Berkeley, Yang Dan lab) to characterise responses to natural images in the cat primary visual cortex, using multivariate nonlinear regression models that I developed ([Rapela et al., 2006, 2010](#)).
- During my first postdoc at UC San Diego, a collaboration with Prof. Edward Chang (UCSF) helped me secure my first seed grant. His lab generously shared outstanding human ECoG recordings from epileptic subjects, which we used to fit and evaluate new ensemble dynamical models that we developed ([Rapela et al., 2015](#)). These rich datasets ultimately enabled me to discover traveling waves in human language regions while subjects pronounced consonant-vowel syllables ([Rapela, 2016, 2017, 2018](#)).
- Also, during my postdoc at UC San Diego, I collaborated with EEG experimental researchers, Prof. Jeanne Townsend and Dr. Marissa Westerfield on the application of investigation of the relation of single-trial phase coherence in the EEG and attentional shifts in humans ([Rapela et al., 2018](#)). The ML innovation of this research was the use of variational inference to estimate large-dimensional linear-regression models from limited amount of data.
- I was hired at the Gatsby Unit to collaborate with SWC experimental neuroscientists on the use of advanced statistical methods to analyse their recordings. One such successful collaboration with Ms. Mitra Javadsadeh No (Hofer lab) used linear dynamical system models, that I proposed and developed, to model long-range cortical communication. She presented orally this research at CoSyNe¹
- At the SWC and Gatsby Unit I contributed my statistical expertise for the analysis of Aeon data ([Campagner et al., 2025](#)). In this Aeon collaboration, I met Dr. Goncalo Lopes, director of NeuroGEARS, with whom I applied, obtained and developed the BBSRC funded project [Machine Intelligence for Neuroscience Experimental Control](#).

Experience of engagement beyond academia, including contributions to policy, practice, public engagement, or knowledge exchange. My career has been deeply committed to knowledge exchange and transforming laboratory practice through the translation of complex machine learning into accessible, open-source scientific tools. Rather than keeping advanced mathematical frameworks confined within theoretical computing, I have focused on bridging the gap between machine learning and its practical, real-world application by experimental scientists.

- **Knowledge Exchange and Community Building:** As a key co-developer of the *Bonsai.ML* ecosystem and the statistical infrastructure for the institute-wide *Aeon* project, I have directly facilitated knowledge exchange across distinct scientific domains. A prominent example of this engagement was my active involvement in the **2024 Bonsai Developers Conference**, where we brought together an international community of developers and experimentalists to share insights, standardize workflows, and lower the barrier to integrating machine learning into local laboratory environments.
- **Transforming Practice via Open Software Distribution:** I have a proven, multi-year track record of developing and openly distributing robust statistical packages to the wider scientific community. By translating complex mathematical frameworks into production-ready software, I have directly enhanced computational workflows across laboratories. My publicly distributed contributions include:
 - [joacorapela/svGPFA](#): Python implementation for Sparse Variational Gaussian Process Factor Analysis.
 - [joacorapela/ssm](#): State-space models tailored for behavioral and kinematic inference.

¹Mitra presented orally our results at CoSyNe 2022. Presentation begins at 2:59:00.

- [joacorapela/hiddenMarkovModels](#): R implementations for HMMs with multivariate Gaussian, Poisson, Poisson GLM, and autoregressive observations.
- [joacorapela/bayesianLinearRegression](#) & [joacorapela/variationalLinearRegression](#): Core computational frameworks for exact and variational Bayesian inference.
- **Contributions to Educational Practice:** I have a proven track record of translating cutting-edge research into educational practice. I designed and taught the foundational *Statistical Neuroscience* course for PhD students at the Sainsbury Wellcome Centre and mentored Simmons Foundation fellows. This work represents a sustained commitment to practice, ensuring that the next generation of biomedical and NeuroAI researchers are equipped with rigorous quantitative skills.

Skills and abilities

Ability to develop and implement team-based solutions to complex challenges, delivering positive outcomes. My entire career, from my PhD studies to my current role at UCL, has been built upon the principles of team science. I firmly believe that the most complex challenges in modern computational neuroscience cannot be solved in isolation; they require the integration of diverse perspectives, technical skill sets, and experimental methodologies. Rather than working as a solitary investigator, I have consistently approached scientific obstacles by developing, leading, and implementing team-based solutions alongside my collaborators.

- **Bridging Theory and Experimentation:** Throughout my seven years at the Gatsby Computational Neuroscience Unit and the Sainsbury Wellcome Centre, I have served as the core collaborative bridge between advanced mathematical theory and complex biological experimentation. Working in continuous partnership with Unit Director Maneesh Sahani, experimental faculty, and engineering teams, I have co-developed institutional software infrastructure like *Bonsai.ML* and the *Aeon* project. These platforms succeeded precisely because they were designed as team-based solutions to the complex challenge of large-scale, automated behavioral and neural data acquisition.
- **Delivering Major Collaborative Outcomes:** This team-first approach has delivered verifiable, positive academic results. I was a primary driver in securing and executing an £800,000 BBSRC grant. Delivering on a grant of this scale required the continuous coordination of cross-disciplinary milestones, aligning data-engineering standards with the active scientific needs of multiple experimental laboratories.
- **A Distributed Mindset to Problem Solving:** From the beginning of my doctoral research to my recent open-source software distributions, every major technical challenge I have overcome has been solved through active, transparent engagement with my peers and mentors. I view software development, pedagogical design, and data analysis not as independent tasks, but as collaborative systems.

As a faculty member, I intend to bring this established culture of team science directly into the joint environment of NPP and CDB. I do not look to build an isolated lab, but rather a collaborative node that actively partners with experimentalists to solve complex neuroscientific challenges through shared, team-based breakthroughs.

Advanced computational skills relevant to neuroscience research, including proficiency in at least one scientific programming language (e.g. Python, MATLAB, R, Julia) and familiarity with machine learning frameworks or data analysis tools used in neuroscience or bioscience. I possess expert-level computational skills developed over more than a decade of research and software engineering in computational neuroscience. I am highly proficient in both Python and R for data analysis, statistical modeling, and scientific visualization. Rather than merely being familiar with existing machine learning frameworks, I actively build within them; I have engineered and optimized the core architecture of the *svGPFA* package utilizing both PyTorch and JAX backbones to handle high-dimensional, real-world neural population data.

UCL Ways of Working

Ability to build collaborative research and industry partnerships. The entirety of my research career in computational neuroscience provides direct evidence of my ability to establish and sustain high-impact collaborative research partnerships. My work naturally exists at the intersection of diverse disciplines, requiring continuous synthesis between theoretical modeling and experimental application.

Beyond traditional academic networks, I have a proven track record of bridging the gap between university research and commercial enterprise. A clear testament to my ability to build industry partnerships is my successful collaboration with **NeuroGEARS Ltd.** Together, we co-developed the strategic vision, successfully bid for funding, and executed the development of the *Bonsai.ML* grant. This industry partnership has been vital in translating advanced machine learning

methods into robust, commercial-grade software ecosystems that are optimized for active use in modern laboratories. As a faculty member at UCL, I intend to leverage these established industry and academic networks to drive innovative, well-funded cross-sector collaborations.

Ability to develop and implement team-based solutions to complex challenges, delivering positive outcomes. Answered above.

Active commitment to equality, diversity, and inclusion, for example through inclusive teaching practices, mentoring underrepresented students, participating in EDI initiatives, or fostering an inclusive lab/group culture. Having been a member of an underrepresented minority in the US and the UK, I am highly sensitive to EDI issues. This was one of my motivations for participating in the SURFiN mentoring program of the Simons Foundation, where I mentored a first student from Pakistan and a second one from China.

I am deeply dedicated to fostering an explicitly inclusive research and classroom culture. In a faculty role at UCL, I will leverage my experiences to implement equitable teaching practices in quantitative courses and actively recruit, support, and advocate for diverse talent within the joint environment of NPP and CDB.

Desirable Criteria

Postgraduate Certificate in Teaching & Learning in Higher Education, or equivalent qualification. While I do not currently hold a formal Postgraduate Certificate in Higher Education, I am highly motivated to obtain this qualification and would actively pursue it upon appointment. I have extensive practical experience in educational design and delivery, having created and taught the foundational *Statistical Neuroscience* course for PhD students at the Sainsbury Wellcome Centre and mentored international fellows. I view the opportunity to complete a formal teaching certificate at UCL as a valuable mechanism to ground my existing pedagogical experience in structured higher-education theory and to ensure my classroom methods remain inclusive and highly effective.

Experience of contributions to peer review bodies/committees, professional organisations, learned societies, government committees or Research Councils etc. I am an ad-hoc reviewer for Vision Research, Frontiers in Neuroscience, and Journal of Perceptual Imaging. In addition, I have reviewed grant proposal for the Software Sustainability Institute.

Experience of an active role in widening participation initiatives that span research communities, e.g. sharing ideas and experience through organising seminars and workshops. I believe that advancing scientific discovery requires building open, transparent, and equitable research environments. To actively contribute to this mission, I served as a member of the Sainsbury Wellcome Centre **Research Culture Working Group (RCWG)**.

In this role, I collaborated with a cross-disciplinary team of researchers and staff to evaluate and systematically improve the research culture of our institute. My contributions focused on fostering a highly collaborative environment that explicitly bridges the gap between theoretical modeling and experimental laboratories. By designing initiatives that encourage the open sharing of data, software workflows, and technical expertise across distinct scientific communities, this work directly advanced widening participation, knowledge equity, and intellectual accessibility within the broader SWC and Gatsby Unit ecosystem.

References

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